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ABSTRACT

Context. Optional, leave empty if necessary. The heading "Context" is used when needed to give background information on the research conducted in the paper
Aims. Mandatory. The objectives of the paper are defined here.
Methods. Mandatory. The methods of the investigation are outlined here
Results. Mandatory. The results are summarized here.
Conclusions. Optional, leave empty if necessary. "Conclusions" can be used to explicit the general conclusions that can be drawn from the paper.

Key words. gravitational waves – strong gravitational lensing – parameter estimation – sky localization – binary black holes – LIGO

1. Introduction

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2. Citations and maths examples

In this section the one-zone model of Baker (1966), originally used to study the Cepheïd pulsation mechanism, will be briefly reviewed, see Fig. 2, Table 4 and Eq. (3). For the one-zone-model Baker obtains necessary conditions for dynamical, secular and vibrational (or pulsational) stability (Eqs. (34a, b, c) in Baker 1966).

$$\tau_{\rm co} = \frac{E_{\rm th}}{L_{r0}}\,, \tag{1}$$

and the *local free-fall time*

$$\tau_{\rm ff} = \sqrt{\frac{3\pi}{32G}\frac{4\pi r_0^3}{3M_{\rm r}}}\,, \tag{2}$$

Baker's K and σ_0 have the following form:

$$\sigma_0 = \frac{\pi}{\sqrt{8}}\frac{1}{\tau_{\rm ff}} \tag{3}$$

$$K = \frac{\sqrt{32}}{\pi}\frac{1}{\delta}\frac{\tau_{\rm ff}}{\tau_{\rm co}}\,; \tag{4}$$

where $E_{\rm th} \approx m(P_0/\rho_0)$ has been used and

$$\begin{aligned}\delta &= -\left(\tfrac{\partial \ln\rho}{\partial \ln T}\right)_P \\ e &= mc^2\end{aligned} \tag{5}$$

is a thermodynamical quantity which is of order 1 and equal to 1 for nonreacting mixtures of classical perfect gases. The physical meaning of σ_0 and K is clearly visible in the equations above. σ_0 represents a frequency of the order one per free-fall time. K is proportional to the ratio of the free-fall time and the cooling time. Substituting into Baker's criteria, using thermodynamic identities and definitions of thermodynamic quantities,

$$\Gamma_1 = \left(\frac{\partial \ln P}{\partial \ln\rho}\right)_S\,,\ \chi_\rho = \left(\frac{\partial \ln P}{\partial \ln\rho}\right)_T\,,\ \kappa_P = \left(\frac{\partial \ln\kappa}{\partial \ln P}\right)_T$$

$$\nabla_{\rm ad} = \left(\frac{\partial \ln T}{\partial \ln P}\right)_S\,,\ \chi_T = \left(\frac{\partial \ln P}{\partial \ln T}\right)_\rho\,,\ \kappa_T = \left(\frac{\partial \ln\kappa}{\partial \ln T}\right)_T$$

3. Figures examples

Examples of figures using graphicx. The guide "Using Imported Graphics in LaTeX2e" by Keith Reckdahl is available on a lot of LaTeXpublic servers or CTAN mirrors.

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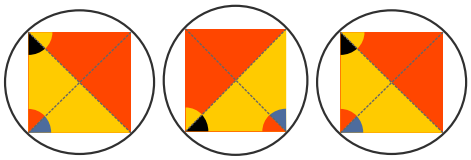


Fig. 1: A onecolumn \figure* with six graphics

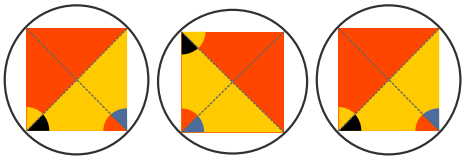


Fig. 1: continued.

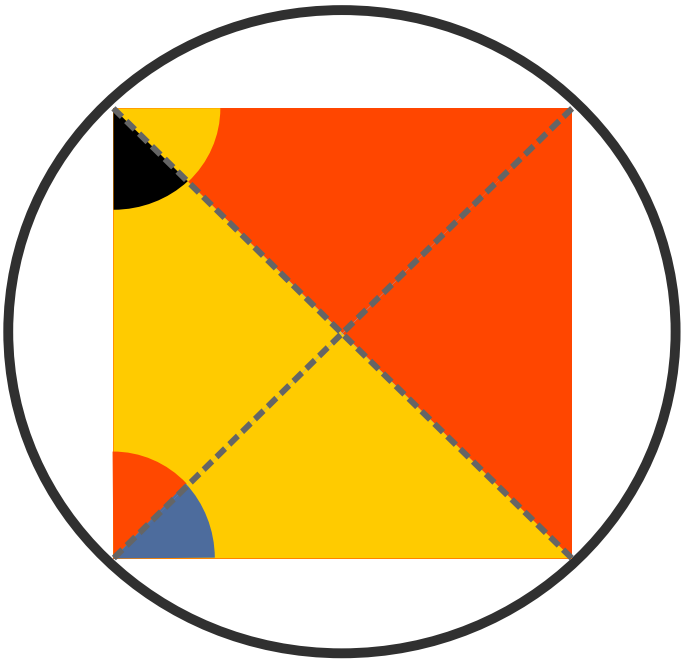


Fig. 2: Figure as large as the column width

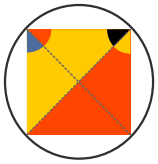


Fig. 3: Rotated figure

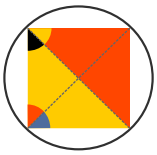


Fig. 4. Figure with caption on the right side

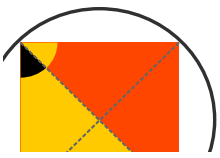


Fig. 5: Figure with a new BoundingBox

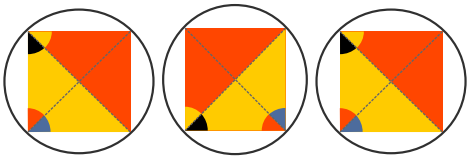


Fig. 6: A figure including three graphics

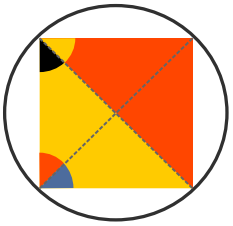


Fig. 7: Continued figure numbering

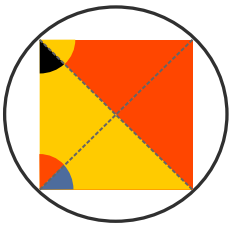


Fig. 7: continued.

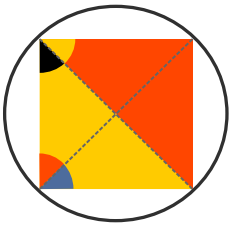


Fig. 7: continued.

4. Tables examples

The jump in table numbering below is caused by the command \longtable*. This command only works in the onecolumn environment. For this reason, we recommend either:

- placing your long tables in onecolumn appendices (cf. D.1 and F.1),
- or using the longtab environment as illustrated by tables 2 and 3. Note that the longtab environment will preserve the table numbering and automatically places long tables after the appendices. They will be moved inside the appendices by the Publisher, if necessary.

Table 1: Simple A&A Table

HJD	E	Method#2	Method#3
1	50	−837	970
2	47	877	230

Table 4: Table with notes

Star	Spectral type	RA(J2000)
69	B1 V	09 15 54.046
LS 1267 (86)	O8 V	09 15 52.787
24.6	7.58	1.37
MO 2-119	B0.5 V	09 15 33.7
LS 1269	O8.5 V	09 15 56.60

Notes. The top panel shows likely members of Pismis 11. The bottom panel displays stars outside the clusters.

Table 5: Table with multiple notes

Star	Spectral type	RA(J2000)
69	B1 V	09 15 54.046
LS 1267 (86)	O8 V	11.07[a]
24.6	7.58[l]	1.37[a]
MO 2-119	B0.5 V	11.74[c]
LS 1269	O8.5 V	10.85[d]

Notes. The top panel shows likely members of Pismis 11. The bottom panel displays stars outside the clusters.
[(a)] Photometry for MF13, LS 1267 and HD 80077 from Dupont et al. [(b)] Photometry for LS 1262, LS 1269 from Durand et al. [(c)] Photometry for MO2-119 from Mathieu et al.

Table 6: Table with references

SN name	Epoch (with respect to B maximum)	Bands
1981B	0	*UBV*
1990N	2, 7	*UBVRI*
1991M	3	*VRI*
SNe 91bg-like		
1991bg	1, 2	*BVRI*
1999by	−5, −4, −3, 3, 4, 5	*UBVRI*
SNe 91T-like		
1991T	−3, 0	*UBVRI*
2000cx	−3, −2, 0, 1, 5	*UBVRI*

References. (1) Zheng (1997); (2) Mizuno (1980); (3) Balluch (1988); (4) Cox (1980); (5) Cox (1969); (6) Tscharnuter (1987); (7) Terlevich (1992); (8) Yorke (1980a).

5. Results: sky localization with H1+L1

We now assess the improvement in sky localization achieved by the coherent joint analysis of strongly lensed image pairs, relative to the analysis of each image in isolation, for the two-detector H1+L1 (HL) network under projected O5a sensitivity, across the full sample of 157 successfully analyzed events

5.1. 90% credible sky area

For each of the 157 events, we compute the 90% credible sky area A_{90} (Eq. (**??**)) from the single-image posteriors of image 1 and image 2, and from the joint posterior. We define the best single-image area as $A_{90}^{\rm best} \equiv \min(A_{90}^{(1)}, A_{90}^{(2)})$, and the per-event improvement factor from combining the two images as

$$\mathrm{IF}_{\rm joint} \equiv A_{90}^{\rm best}/A_{90}^{\rm joint}. \tag{6}$$

Figure 8 shows the cumulative and probability density distributions of A_{90} for the best single-image and joint analyses. The joint distribution is shifted to smaller areas across the entire sample: the median 90% credible area drops from 562.5 deg^2 (best single image) to 90.7 deg^2 (joint), a median improvement factor of $\mathrm{IF}_{\rm joint} = 5.19$. The improvement is universal, with $\mathrm{IF}_{\rm joint} > 1$ for all 157 events (range 1.11–27.90), confirming that the gain is systematic rather than driven by a handful of favorably placed sources. Six events (3.8 %) reach a joint area below 10 deg^2, the benchmark commonly adopted for host-galaxy identification and multimessenger follow-up with wide-field survey telescopes (**?**).

5.2. Dependence on network SNR

Figure 9 shows A_{90} as a function of the per-image network SNR, for each image and for the joint analysis. The single-image areas show substantial scatter at fixed SNR, reflecting the dominant role of sky position and antenna pattern orientation, rather than SNR alone, in setting the localization precision of a two-detector network (§**??**). The joint analysis consistently reduces A_{90} relative to either single image, with the largest gains occurring for events whose single-image areas are largest to begin with.

5.3. Constellation localization

We further characterize the localization using the fraction of posterior probability contained in the dominant IAU constellation

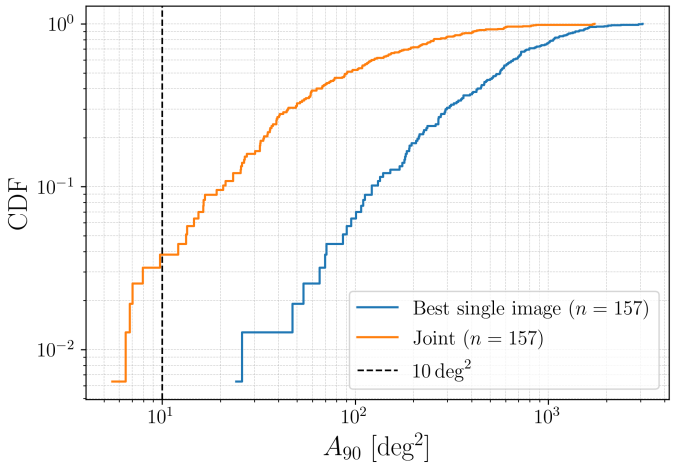


Fig. 8: Cumulative (left) and probability density (right) distributions of the 90% credible sky area, for the best single-image analysis and the joint analysis, across the 157 analyzed events. The vertical dashed line marks 10 deg^2.

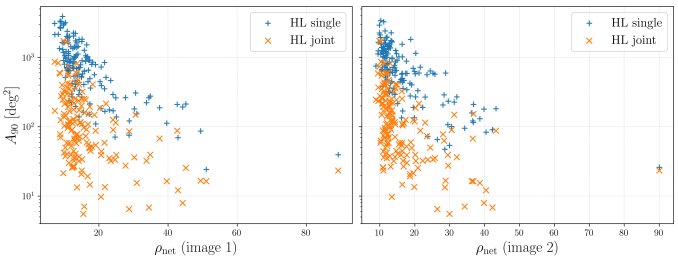


Fig. 9: 90% credible sky area versus network SNR for image 1 (left) and image 2 (right). Blue crosses: single-image analysis; orange crosses: joint analysis for the same event.

(**?**). For single-image analyses the median fraction in the top constellation is 41.75 % (best of the two images). The joint analysis raises the median to 69.3 %, with 121/157 events (77.1 %) above 50 % and 49/157 events (31.2 %) above 90 %.

Comparing against the published sky maps of 104 GWTC-5 events (**?**), we find that the joint lensed-pair analysis with only two detectors reaches constellation fractions comparable to those achieved by the three-detector HLV network on real data, whereas real HL-only events are confined below ∼ 50 % (Fig. 10). This places the improvement from the joint lensed analysis on a comparable footing to the localization gain from adding a third detector, using only two detectors and a repeated observation of the same source, a connection we examine directly in §6.

5.4. Comparison with previous work

? demonstrated that quadruply lensed events observed by the full LIGO/Virgo/KAGRA network can, combined with electromagnetic constraints on the lens galaxy, be localized to one or a few candidate host galaxies. **?** performed the first systematic study of repeated-signal localization using BAYESTAR sky maps on a simulated population observed with the three-detector HLV network, finding roughly an order-of-magnitude reduction in the 90% credible area from combining two images. The smaller median improvement factor found here (5.19) is consistent with the more conservative two-detector configuration adopted in this

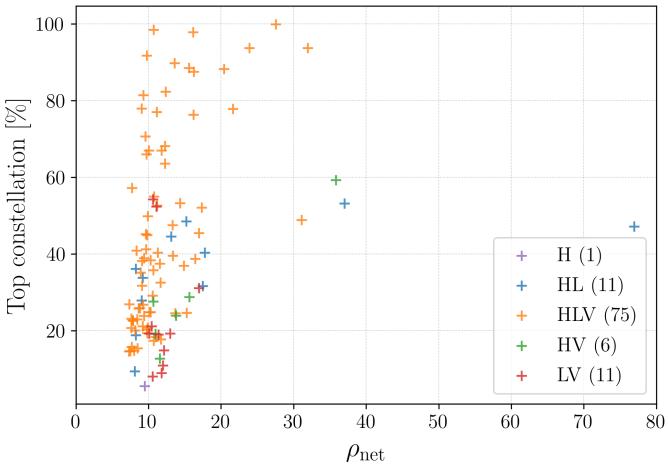


Fig. 10: Probability fraction in the dominant constellation versus network SNR for 104 GWTC-5 events, colored by detector network. The HL subset, matching the detector configuration used in this work, is confined below ∼ 50 %.

work: with only H1 and L1, single-image localization already relies more heavily on antenna pattern information than on triangulation, leaving comparatively less room for a second image, which acts through the same antenna-pattern channel, to improve upon it. Both studies nonetheless converge on the same qualitative conclusion, namely that combining two lensed images yields a substantial and consistent sky localization gain regardless of detector network or analysis method. §6 tests this conclusion directly by asking how the gain from the lensed pair interacts with growing the detector network itself.

6. Adding a third detector: Virgo

The results of §5 isolate the gain from a coherent joint analysis of two lensed images within a fixed two-detector network. We now ask a complementary question: how much further does sky localization improve when a third detector, Virgo, is added to the network, and does that gain depend on the geometry of the event relative to the H1-L1 baseline?

6.1. A stratified subsample, not a random one

Re-running the full 157-event population with Virgo added is computationally prohibitive within the scope of this work. We instead select a stratified subsample of 35 events, designed to probe four distinct regimes of expected Virgo benefit, rather than drawing a random subset: `core` (the 15 events with the largest fractional SNR gain from adding V1, computed with the exact inner-product SNR estimator of GWSNR), `geom` (the 10 events for which neither lensed image is well aligned with the H1-L1 baseline, quantified by the angle between the source direction and the baseline at each image's arrival time, so that triangulation with H1+L1 alone is least reliable), `control` (5 events with low expected SNR gain and good H1-L1 alignment, included as a null case), and `threshold` (5 events whose H1+L1 joint area sits close to the 10 deg^2 benchmark of §5.1, where a modest absolute change in area has an outsized practical effect on follow-up feasibility).

Because this selection deliberately overrepresents events favorable to Virgo, the median statistics reported below describe this stratified subsample and *must not* be compared directly to

the population-level medians of §5. Instead, every comparison in this section is made event by event: the same 35 events are re-analyzed with the H1+L1+V1 (HLV) network, using the Virgo O5 low-sensitivity design curve (**?**), under the same pipeline, priors, and waveform model as the H1+L1 (HL) analysis, and the two outcomes for each event are compared directly. The full per-event table is given in Appendix A.

6.2. Sky area: HL versus HLV

Figure 11 shows the joint 90% credible area with HLV against the same quantity with HL, for all 35 events. Every event falls below the diagonal: adding Virgo improves the localization of every event in the subsample, with no exceptions. The per-event improvement factor, $\mathrm{IF}_{\mathrm{Virgo}} \equiv A_{90}^{\mathrm{joint,\,HL}}/A_{90}^{\mathrm{joint,\,HLV}}$, has a median of 13.3 across the subsample (range 2.4–93.8), and the median joint area drops from 71.2 deg^2 (HL) to 8.8 deg^2 (HLV). Twenty of the 35 events (57.1 %) fall below the 10 deg^2 benchmark with Virgo, compared to a single event with H1+L1 alone. The median dominant-constellation fraction rises correspondingly, from 73.3 % (HL) to 99.5 % (HLV).

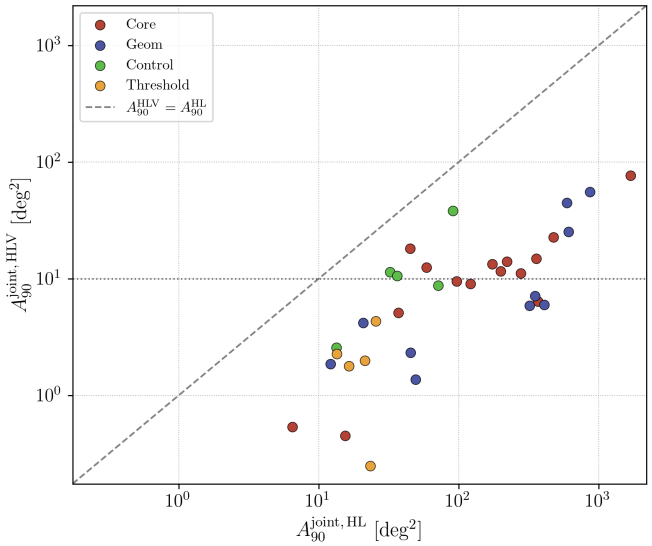


Fig. 11: Joint 90% credible area with H1+L1+V1 (HLV) versus H1+L1 (HL), for the 35-event stratified subsample, colored by selection group. The dashed line marks equal area; every point falls below it. The dotted line marks 10 deg^2.

Figure 12 shows the same comparison per event, as a slope chart grouped by selection category, making the heterogeneity of the gain across groups directly visible: the `core` and `geom` groups, selected respectively for high expected SNR gain and for poor H1-L1 geometric alignment, show the largest improvements (median $\mathrm{IF}_{\mathrm{Virgo}} = 15.8$ and 21.9), while the `control` group, selected as a null case of low SNR gain and favorable H1-L1 alignment, shows the smallest (median 3.4). The `threshold` group, selected independently of either criterion, falls in between (median 9.2). This ordering confirms that the selection criteria used to build the subsample track the physical drivers of the Virgo gain: events that are geometrically degenerate under H1+L1, or that happen to be loud in Virgo, benefit the most, while events already well constrained by H1+L1 triangulation gain comparatively little from a third detector.

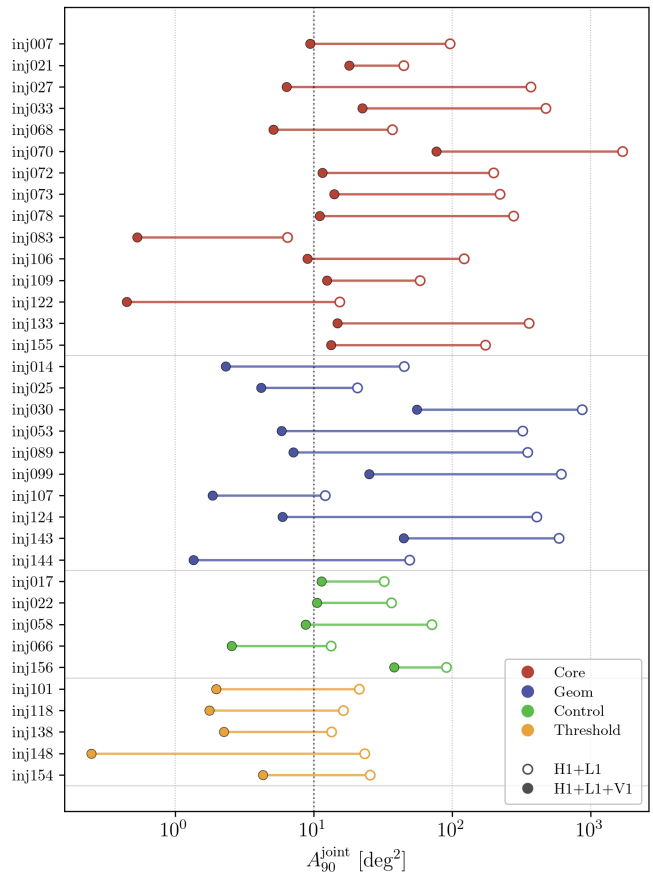


Fig. 12: Joint 90% credible area for each of the 35 events, with H1+L1 (open markers) and H1+L1+V1 (filled markers), grouped by selection category. The dotted line marks 10 deg^2.

6.3. The diminishing, but still positive, value of the lensed pair

A natural question is whether the gain from combining two lensed images, the central result of §5, persists once a third detector is already available. Within the HL network, the median value of $\mathrm{IF}_{\mathrm{joint}}$ (Eq. (6)) for this 35-event subsample is 4.68, consistent with the 5.19 found for the full 157-event population (§5.1) and confirming that the subsample is not biased with respect to this particular metric. Within the HLV network, the same quantity, which we denote $\mathrm{IF}_{\mathrm{joint}}^{\mathrm{HLV}}$ to distinguish it from its HL counterpart $\mathrm{IF}_{\mathrm{joint}}^{\mathrm{HL}}$, drops to 2.20.

This does not mean that combining the two lensed images becomes uninformative once Virgo is added. Every event in the subsample still benefits from the joint analysis within the HLV network itself (Appendix A), and the absolute localization achieved by the HLV joint analysis remains, by a wide margin, the best result available for any of these events. What the drop in $\mathrm{IF}_{\mathrm{joint}}$ indicates is that the specific role played by the second lensed image, namely supplying an independent triangulation baseline through its different arrival time and antenna-pattern orientation, is partially subsumed once a third detector already provides that information directly. The lensed-pair technique and a richer detector network are therefore complementary rather than redundant: the former matters most precisely in the regime probed by the `geom` group, where the detector network alone is geometrically degenerate, and its marginal contribution

shrinks, without vanishing, as the network's own triangulation power increases.

6.4. Comparison with the literature

These results extend the central finding of §5.4 along a second, complementary axis: while §5 shows that repeating an observation through a lensed image pair substantially improves localization within a fixed detector network, the present section shows that growing the detector network itself delivers a comparable, and in the most favorable cases much larger, improvement, and that the two strategies interact in the physically expected way, each compensating most for the specific geometric limitations the other leaves unaddressed.

7. Conclusions

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Acknowledgements. Part of this work was supported by *ESO*, project number Ts 17/2–1.

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Appendix A: Full per-event results for the Virgo subsample

Table A.1 lists, for each of the 35 events analyzed with H1+L1+V1 (§6), the selection group, the joint 90% credible area with H1+L1 (HL) and H1+L1+V1 (HLV), the per-event improvement factor from adding Virgo to the joint analysis ($\mathrm{IF}_{\mathrm{Virgo}}$), and the improvement factor from combining the two lensed images within each network separately ($\mathrm{IF}^{\mathrm{HL}}_{\mathrm{joint}}$, $\mathrm{IF}^{\mathrm{HLV}}_{\mathrm{joint}}$), as discussed in §6.3.

Table A.1: Per-event sky-area results for the 35-event Virgo subsample.

Event	Group	$A^{\mathrm{joint,\,HL}}_{90}$ [deg^2]	$A^{\mathrm{joint,\,HLV}}_{90}$ [deg^2]	$\mathrm{IF}_{\mathrm{Virgo}}$	$\mathrm{IF}^{\mathrm{HL}}_{\mathrm{joint}}$	$\mathrm{IF}^{\mathrm{HLV}}_{\mathrm{joint}}$
inj027	Core	367.8	6.4	57.64	1.89	2.20
inj122	Core	15.4	0.4	34.36	8.48	0.68
inj078	Core	276.9	11.1	25.02	7.66	2.69
inj133	Core	357.3	14.8	24.07	1.70	2.26
inj070	Core	1690.2	76.7	22.03	1.41	2.15
inj033	Core	473.0	22.5	20.98	3.06	1.78
inj072	Core	199.4	11.6	17.19	2.68	2.37
inj073	Core	221.3	14.0	15.76	3.06	1.70
inj106	Core	121.4	9.0	13.48	6.98	1.31
inj155	Core	173.7	13.3	13.03	5.51	1.29
inj083	Core	6.5	0.5	12.14	10.09	2.14
inj007	Core	96.6	9.4	10.22	10.12	1.47
inj068	Core	36.9	5.1	7.23	3.78	2.67
inj109	Core	58.8	12.4	4.72	4.03	2.29
inj021	Core	44.7	18.1	2.47	11.15	1.58
inj124	Geom	407.9	5.9	68.61	1.77	2.74
inj053	Geom	321.2	5.9	54.76	1.80	1.58
inj089	Geom	350.5	7.1	49.23	3.07	2.43
inj144	Geom	49.2	1.4	36.21	8.18	2.45
inj099	Geom	609.7	25.1	24.28	2.14	2.93
inj014	Geom	45.1	2.3	19.42	8.05	1.83
inj030	Geom	865.1	55.5	15.58	1.85	1.59
inj143	Geom	590.9	44.5	13.27	2.02	1.38
inj107	Geom	12.1	1.9	6.53	5.76	1.81
inj025	Geom	20.7	4.2	4.96	13.42	2.35
inj058	Control	71.2	8.8	8.13	4.71	6.05
inj066	Control	13.3	2.6	5.22	3.55	6.47
inj022	Control	36.3	10.6	3.42	7.42	3.93
inj017	Control	32.3	11.4	2.84	5.91	2.26
inj156	Control	90.7	38.2	2.38	4.68	2.93
inj148	Threshold	23.3	0.2	93.77	1.11	1.65
inj101	Threshold	21.3	2.0	10.79	8.92	1.29
inj118	Threshold	16.4	1.8	9.24	1.48	2.75
inj138	Threshold	13.5	2.3	5.96	10.00	1.93
inj154	Threshold	25.6	4.3	5.95	10.58	4.01

Appendix B: Wide tables and figures after an appendix title: recommended method

In the PDF output, floats should be placed under their own appendix, not before the title, nor after the title of the next appendix. In short appendices, one-column floats {figure*} or {table*} will generate a blank page. To prevent this behaviour, we recommend to switch to \onecolumn and set the [ht!] parameter in your floats: please check the LaTeXcode of this appendix.

In case you have a lot of floating objects for little text and the LaTeXengine moves the floats away from their context, the command \FloatBarrier of the "placeins" package will empty the float buffer and place all stored floats in the continuity. If you still encounter problems with wide floats placement, just use the \onecolumn environment throughout the appendices.

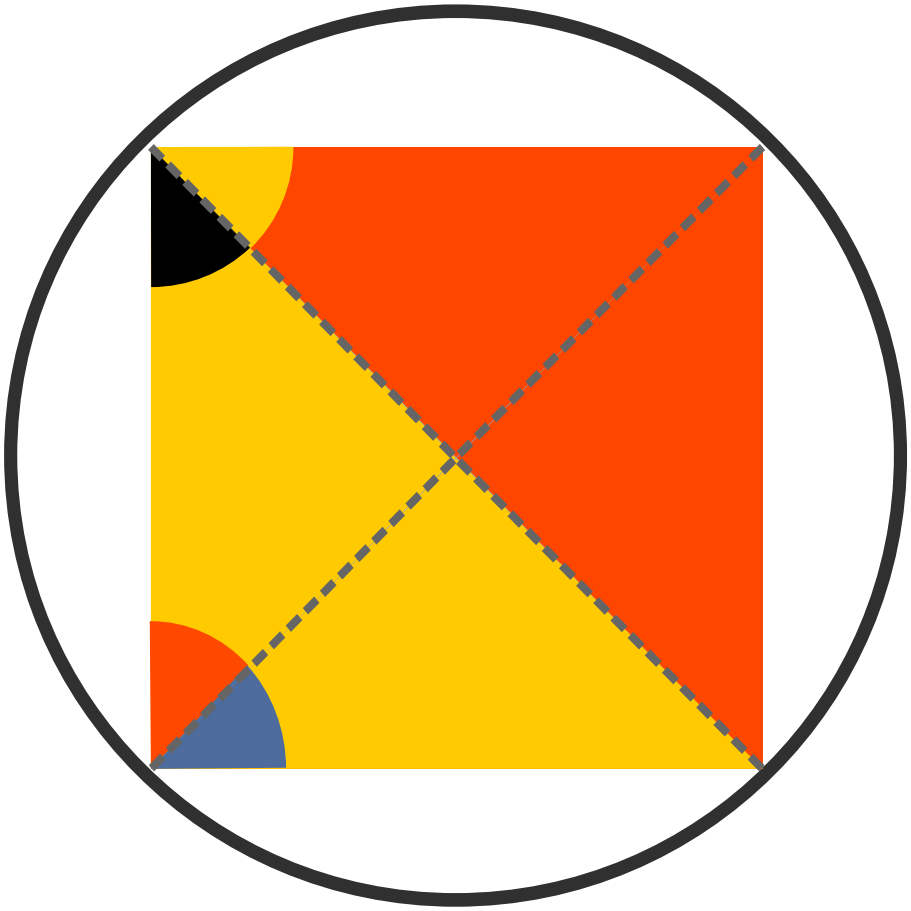


Fig. B.1: A one-column {figure*}[ht!] after a section title. If text follows like below, it is easier to finish the section in \onecolumn. If needed, you may revert to \twocolumn when reaching the next page.

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

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Appendix C: Wide tables and figures after an appendix title: alternate method

To prevent a blank page, a second method is to insert the appendix title after declaring the onecolumn float.
This method should be reserved to appendices containing only one-column floats{figure*} or {table*} and no text.

Table C.1: A one-column {table*}

ISO-L1551	$F_{6.7}$ [mJy]	$\alpha_{6.7-14.3}$	YSO type[d]	Status	Comments
New YSO candidates					
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
Previously known YSOs					
61	0.89 ± 0.58	1.77	Class I	HH 30	Circumstellar disk
96	38.34 ± 0.71	37.5	Class II	MHO 5	Spectral type

Appendix D: Long tables in appendices

For long tables (multipage) in appendices, we use the method described in appendix A. For long landscape tables, please refer to Appendix E.

Table D.1: A long table

Catalogue	M_V	Spectral	Distance	Mode	Count Rate
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86 [1]	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230

[1] Source not included in the HRI catalog. See Sect. 5.4.2 for details.

Table D.1: continued.

Catalogue	M_V	Spectral	Distance	Mode	Count Rate
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230

Appendix E: Rotated single page tables

To prevent a blank page with {sidewaystable*}, we use the method described in appendix B: declare the table first, and the section second.

Table E.1: A rotated table with {sidewaystable*}

ISO-L1551	$F_{6.7}$ [mJy]	$\alpha_{6.7-14.3}$	YSO type[d]	Status	Comments
		New YSO candidates			
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
1	1.56 ± 0.47	–	Class II[c]	New	Mid
2	0.79:	0.97:	Class II ?	New	
3	4.95 ± 0.68	3.18	Class II / III	New	
5	1.44 ± 0.33	1.88	Class II	New	
		Previously known YSOs			
61	0.89 ± 0.58	1.77	Class I	HH 30	Circumstellar disk
96	38.34 ± 0.71	37.5	Class II	MHO 5	Spectral type

Appendix F: Rotated long tables in appendices

For rotated long tables in appendices, we use the method described in appendix A, combined with {landscape}.

Table F.1: A long landscape table

Catalogue	M_V	Spectral	Distance	Mode	Count Rate
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86 [2]	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686

[2] Source not included in the HRI catalog. See Sect. 5.4.2 for details.

Table F.1: continued.

Catalogue	M_V	Spectral	Distance	Mode	Count Rate
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230

Table 2: A long table using the longtab environment

Catalogue	M_V	Spectral	Distance	Mode	Count Rate
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86 [3]	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686

[3] Source not included in the HRI catalog. See Sect. 5.4.2 for details.

Table 2: continued.

Catalogue	M_V	Spectral	Distance	Mode	Count Rate
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230

Table 3: A long landscape table using the longtab environment

Catalogue	M_V	Spectral	Distance	Mode	Count Rate
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86 [4]	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170

[4] Source not included in the HRI catalog. See Sect. 5.4.2 for details.

Table 3: continued.

Catalogue	M_V	Spectral	Distance	Mode	Count Rate
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230
Gl 33	6.37	K2 V	7.46	S	0.043170
Gl 66AB	6.26	K2 V	8.15	S	0.260478
Gl 68	5.87	K1 V	7.47	P	0.026610
				H	0.008686
Gl 86	5.92	K0 V	10.91	S	0.058230